



```
In [1]: def greet():  
        print('jyothi')
```

```
In [2]: def greet(): #greet) is the function name -- declaration  
        print('hello good morning')  
        greet() # call the function
```

hello good morning

```
In [3]: def greet():  
        print('hello good morning')  
        greet()  
  
        def greet(): #greet) is the function name -- declaration  
            print('hello good morning')  
            greet()  
  
        def greet(): #greet) is the function name -- declaration  
            print('hello good morning')  
            greet()
```

hello good morning
hello good morning
hello good morning

```
In [4]: def greet():  
        print('hello good morning')  
  
        greet()  
        greet()  
        greet()
```

hello good morning
hello good morning
hello good morning

```
In [5]: def add(x,y): #x & y -- formal argument  
        c=x+y  
        print(c)  
        add(5,6) # 5 & 6 is called actual argument
```

11

```
In [6]: def add(x,y):  
        c = x + y  
        print(c)  
  
        r = add(5,6)  
        print(r)  
        print(type(r))
```

11
None
<class 'NoneType'>

```
In [7]: def add(x,y):  
        c = x + y  
        return c  
  
        r = add(5,6)  
        print(r)  
        print(type(r))
```

```
11  
<class 'int'>
```

```
In [8]: def add(x,y):  
        c = x + y  
        return c  
  
        add(3,4,5)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[8], line 5  
      2     c = x + y  
      3     return c  
----> 5 add(3,4,5)  
  
TypeError: add() takes 2 positional arguments but 3 were given
```

```
In [9]: def add(x,y,z):  
        c = x + y + z  
        return c  
  
        add(3,4,5)
```

Out[9]: 12

```
In [10]: def add(x,y,z):  
         c = x + y + z  
         return c  
  
         add(3,4)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[10], line 5  
      2     c = x + y + z  
      3     return c  
----> 5 add(3,4)  
  
TypeError: add() missing 1 required positional argument: 'z'
```

```
In [11]: def add(x, y):  
         c = x + y  
         return c  
         print(add(3,4))
```

```
def greet():  
    return 'hello good morning'  
print(greet())
```

7
hello good morning

```
In [12]: def add(x, y):  
         c = x + y  
         return c  
  
         def greet():  
             return 'hello good morning'  
  
         print(add(3,4))  
         print(greet())
```

7
hello good morning

```
In [13]: def add(x, y):  
         c = x + y  
         return c  
  
         def greet():  
             return 'hello good morning'  
  
         r1 = add(3,4)  
         r2 = greet()  
  
         print(r1)  
         print(r2)  
  
         print(type(r1))  
         print(type(r2))
```

7
hello good morning
<class 'int'>
<class 'str'>

```
In [14]: def add(x, y):  
         c = x + y  
         print(c)  
  
         def greet():  
             print('hello good morning')  
  
         r1 = add(3,4)  
         r2 = greet()  
  
         print(r1)  
         print(r2)  
  
         print(type(r1))
```

```
print(type(r2))
```

```
7
hello good morning
None
None
<class 'NoneType'>
<class 'NoneType'>
```

```
In [15]: def greet():
          print('jyothi')
          print('good noon')

          def add(x,y):
              c=x+y
              print(c)

          def sub(x,y,z):
              d = x-y-z
              print(d)

          greet()
          add(5,4)
          sub(12,4,6)
```

```
jyothi
good noon
9
2
```

```
In [16]: def greet():
          print('jyothi')

          def add_sub(x,y):
              c=x+y
              d=x-y
              print(c)
              print(d)

          greet()
          add_sub(7,2)
```

```
jyothi
9
5
```

```
In [17]: def greet():
          print('jyothi')

          def add_sub(x,y):
              c=x+y
              d=x-y
              print(c)
              print(d)
```

```
r1 = greet()
r2 = add_sub(7,2)

print(type(r1))
print(type(r2))
```

```
jyothi
9
5
<class 'NoneType'>
<class 'NoneType'>
```

```
In [18]: def greet():
         return 'jyothi'

         def add_sub(x,y):
             c=x+y
             d=x-y
             return c, d

         r1 = greet()
         r2 = add_sub(7,2)

         print(r1)
         print(r2)

         print(type(r1))
         print(type(r2))
```

```
jyothi
(9, 5)
<class 'str'>
<class 'tuple'>
```

```
In [19]: def greet():
         return 'jyothi'

         def add_sub(x,y):
             c=x+y
             d=x-y
             return c
             return d

         r1 = greet()
         r2 = add_sub(7,2)

         print(r1)
         print(r2)

         print(type(r1))
         print(type(r2))
```

```
jyothi
9
<class 'str'>
<class 'int'>
```

```
In [20]: def greet():
          return 'jyothi'

          def add_sub(x,y):
              c=x+y
              d=x-y
              return c,d

          r1 = greet()
          r2,r3 = add_sub(5,4)

          print(r1)
          print(r2)
          print(r3)

          print(type(r1))
          print(type(r2))
          print(type(r3))
```

```
jyothi
9
1
<class 'str'>
<class 'int'>
<class 'int'>
```

```
In [21]: def add(a,b ): # a & b called formal argument
          c = a+b
          print(c)

          add(5,6 ) # 5 & 6 actual argument
```

11

1- positional argument

```
In [22]: def person(name,age):
          print(name)
          print(age)

          person('nit')
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[22], line 5
      2     print(name)
      3     print(age)
----> 5 person( )

TypeError: person() missing 1 required positional argument: 'age'
```

```
In [23]: def person(name):
          print(name)
          print(age)

          person('nit', 23)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[23], line 5
      2     print(name)
      3     print(age)
----> 5 person( , 23)

TypeError: person() takes 1 positional argument but 2 were given
```

```
In [24]: def person(name, age):
          print(name)
          print(age)

          person('jyothi', 23)
```

```
jyothi
23
```

```
In [25]: def person(age, name):
          print(name)
          print(age)

          person('jyothi', 23)
```

```
23
jyothi
```

```
In [26]: def person(age, name):
          print(name)
          print(age+1)

          person('jyothi', 23)
```

```
23
```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[26], line 5
      2     print(name)
      3     print(age+1)
----> 5 person(          , 23)

Cell In[26], line 3, in person(age, name)
      1 def person(age, name):
      2     print(name)
----> 3     print(age+1)

TypeError: can only concatenate str (not "int") to str

```

2- Keyword Argument

```

In [27]: def person(age, name):
          print(name)
          print(age+1)

          person(name = 'jyothi', age = 23)

```

```

jyothi
24

```

```

In [28]: def person(age, name):
          print(name)
          print(age+1)

          person('jyothi', 23)

```

```

23

```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[28], line 5
      2     print(name)
      3     print(age+1)
----> 5 person(          , 23)

Cell In[28], line 3, in person(age, name)
      1 def person(age, name):
      2     print(name)
----> 3     print(age+1)

TypeError: can only concatenate str (not "int") to str

```

```

In [29]: def person(age, name, mob):
          print(name)
          print(age+1)
          print(mob)

          person(name = 'jyothi', age = 23, mob = 12345)

```



```
jyothi
24
12345
```

```
In [30]: def person(age, name, mob):
          print(name)
          print(age+1)
          print(mob)

          person(name = 'jyothi', age = 23, 12345)
```

```
Cell In[30], line 6
    person(name = 'jyothi', age = 23, 12345)
                                         ^
SyntaxError: positional argument follows keyword argument
```

default argument

```
In [31]: def person(name, age):
          print(name)
          print(age)

          person('jyothi')
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[31], line 5
      2     print(name)
      3     print(age)
----> 5 person( )

TypeError: person() missing 1 required positional argument: 'age'
```

```
In [32]: def person(name, age=18):
          print(name)
          print(age)

          person('jyothi')
```

```
jyothi
18
```

```
In [33]: def person(name, age=20):
          print(name)
          print(age)

          person('jyothi')
```

```
jyothi
20
```

variable length argument

```
In [34]: def sum(a, b):  
         c = a+b  
         print(c)  
  
         sum(5,6)
```

11

```
In [35]: def sum(a, b):  
         c = a+b  
         print(c)  
  
         sum(5,6,7,8)
```

```
-----  
TypeError                                 Traceback (most recent call last)  
Cell In[35], line 5  
      2     c = a+b  
      3     print(c)  
----> 5 sum(5,6,7,8)  
  
TypeError: sum() takes 2 positional arguments but 4 were given
```

```
In [36]: def sum(a, *b): # 1st argument is fixed but for 2nd argument  
  
         print(type(a))  
         print(type(b))  
  
         print(a)  
         print(b)  
  
         sum(5,6,7,8)
```

```
<class 'int'>  
<class 'tuple'>  
5  
(6, 7, 8)
```

```
In [37]: def sum(a, *b): # 1st argument is fixed & we fetch each value from the tuple &  
         c = a  
  
         for i in b:  
             c = c + i  
         print(c)  
  
         sum(5,6,7,8)
```

26

```
In [38]: def sum(a, *b): # 1st argument is fixed & we fetch each value from the tuple &  
         c = a
```

```
    for i in b:
        c = c + i
    print(c)

sum(5,6,7,8, 9, 10, -1, -2, 100)
```

142

```
In [39]: # KWARGS ( KW + ARGS )

def person(name,*data):
    print(name)
    print(data)

person('ALEX', 36, 'JOHN', 987767)
```

ALEX
(36, 'JOHN', 987767)

```
In [40]: def person(name,*data):
    print(name)
    print(data)

    person('ALEX', 36, 'JOHN', 987767)
```

ALEX
(36, 'JOHN', 987767)

```
In [41]: def person(name, **data):
    print('name')
    print(data)

    person('ALEX', age = 36, home_place = 'southcity', mob = 987767)
```

name
{'age': 36, 'home_place': 'southcity', 'mob': 987767}